

REIGN: A Composite Metric for Quantifying NBA Player Impact Across Eras

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Abstract

We present REIGN (Relative Era-adjusted Impact Gauge for NBA players), a composite metric for cross-era player comparison spanning 80 years (1946–2025). REIGN employs four era-specific models, a 60/40 dampened blend for Modern era recomputation, and opponent-adjusted playoff evaluation. Covers 29,969 player-seasons across 3,484 players with both RS and playoff modes.

1. Introduction

Comparing NBA players across eras is complicated by systematic differences in pace, rules, and data availability. REIGN addresses this by defining era-specific models using the best available data in each period, calibrated to produce comparable output scales through era normalization.

2. Data

29,969 player-season observations from 1946–25. Modern era enriched with 8,710 scraped advanced stat records from Basketball Reference (81 international names required manual encoding fixes).

Era	Years	Seasons	Key Stats	R ²
Pioneer	1946–62	1,776	PTS, REB, AST, TS%, WS, PER	0.916
Legacy	1963–95	6,658	+ BPM, VORP, WS/48, STL, BLK	0.917
Classic	1996–2012	5,012	+ 3P%, full advanced	0.891
Modern	2013–25	4,211	Full advanced (enriched)	0.790

Table 1: Data and model fit by era.

3. Methodology

3.1 Core Decomposition

$$\text{REIGN} = \text{REIGN}_{\text{OFF}} + \text{REIGN}_{\text{DEF}}$$

3.2 Era-Specific Models

Pioneer: $\text{REIGN}_P \approx 26.7 \cdot \text{TS\%} - 9.0 \cdot \text{FG\%} + 0.61 \cdot \text{AST} + 0.58 \cdot \text{DWS}$ [R²=0.916]

Legacy: $\text{REIGN}_L \approx 26.2 \cdot \text{WS/48} + 1.27 \cdot \text{VORP} + 1.04 \cdot \text{OWS} + 0.98 \cdot \text{DWS} + 3.2 \cdot \text{TS\%}$ [R²=0.917]

Classic: $\text{REIGN}_C \approx 21.4 \cdot \text{WS/48} + 2.32 \cdot \text{VORP} + 11.4 \cdot \text{TS\%} + 1.51 \cdot \text{STL}$ [R²=0.891]

Modern: $\text{REIGN}_M \approx 47.8 \cdot \text{WS/48} + 1.89 \cdot \text{VORP} + 3.63 \cdot \text{STL} + 2.55 \cdot \text{BLK}$ [R²=0.790]

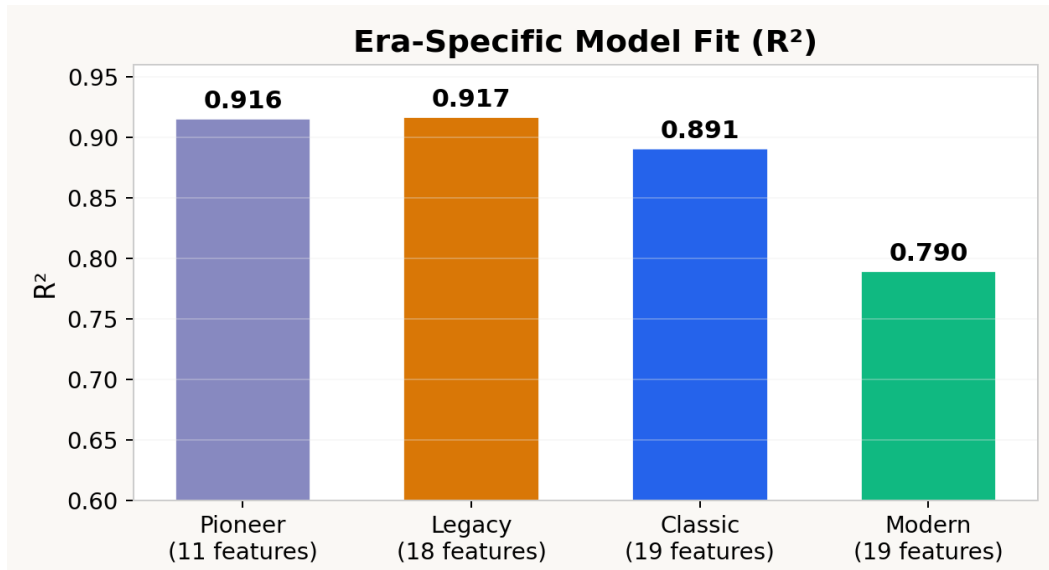


Figure 1: Era-specific model R^2 comparison. All models use WS/48 as dominant predictor except Pioneer (TS%).

3.3 Dampened Blend (Modern Era)

$$\text{REIGN}_{\text{final}} = 0.6 \cdot \text{REIGN}_{\text{original}} + 0.4 \cdot \text{REIGN}_{\text{new model}}$$

New predictions are calibrated to match the original distribution before blending. Max change: ± 4.7 points. Applied to 3,732 RS and 1,606 PO Modern era seasons. PO scores capped at 130% of RS peak.

3.4 Era Normalization

$$X^* = (X - \mu_{\text{window}}) / \sigma_{\text{window}}$$

Z-score within rolling 5-year windows ensures +20 REIGN represents the same relative dominance across all eras.

3.5 Playoff Opponent Strength Adjustment

$$\text{OppFactor} = 0.5 + 0.5 \cdot (Q_{\text{opponents}} / Q_{\text{avg}}) \quad \text{REIGN}_{\text{adj}} = \text{REIGN}_{\text{raw}} \times \text{OppFactor}$$

Constrains adjustments to $\pm 4\%$. Mean factor 0.999, $\sigma=0.010$ over 8,792 playoff seasons.

4. Results

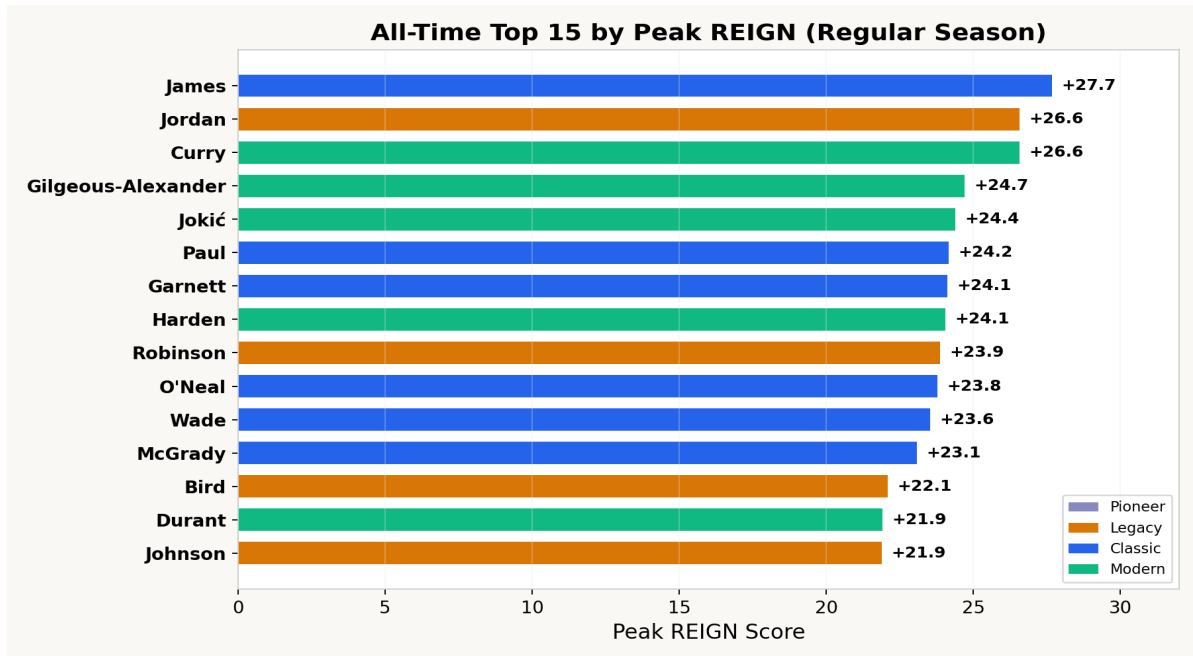


Figure 2: All-time top 15 by peak REIGN (regular season). Colors indicate primary era.

4.1 Regular Season Rankings

#	Player	Peak	3yr	5yr	Era(s)
1	LeBron James	+27.70	+27.02	+25.89	Cla/Mod
2	Michael Jordan	+26.59	+25.73	+25.25	Leg/Cla
3	Stephen Curry	+26.59	+23.23	+21.59	Modern
4	S. Gilgeous-Alexander	+24.72	+23.07	—	Modern
5	Nikola Jokić	+24.40	+22.46	+20.60	Modern
6	Chris Paul	+24.18	+22.40	+21.09	Cla/Mod
7	Kevin Garnett	+24.13	+21.92	+20.66	Leg/Cla
8	James Harden	+24.06	+23.42	+21.15	Modern
9	David Robinson	+23.88	+22.20	+21.28	Leg/Cla
10	Shaquille O'Neal	+23.81	+23.00	+21.88	Leg/Cla

Table 2: Top 10 by peak REIGN (RS).

4.2 Playoff Rankings

#	Player	Peak PO	3yr PO	5yr PO
1	Kawhi Leonard	+25.95	+22.33	+19.61
2	LeBron James	+25.64	+24.88	+24.36
3	Michael Jordan	+25.28	+23.76	+22.59

4	Hakeem Olajuwon	+24.22	+22.40	+19.85
5	James Harden	+23.05	+22.02	—

Table 3: Top 5 by peak playoff REIGN.

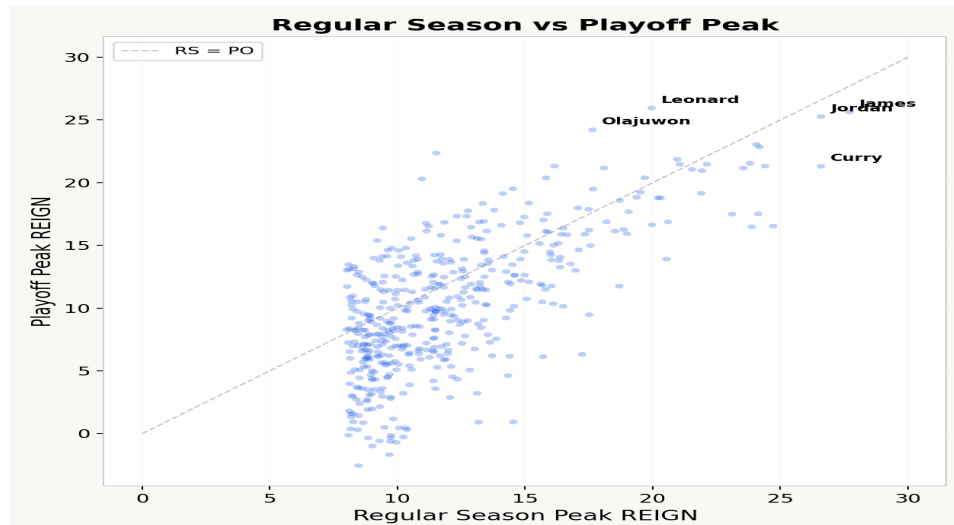


Figure 3: Regular season vs playoff peak REIGN. Dashed line = parity. Players above the line elevate in playoffs.

4.3 Cross-Era Analysis

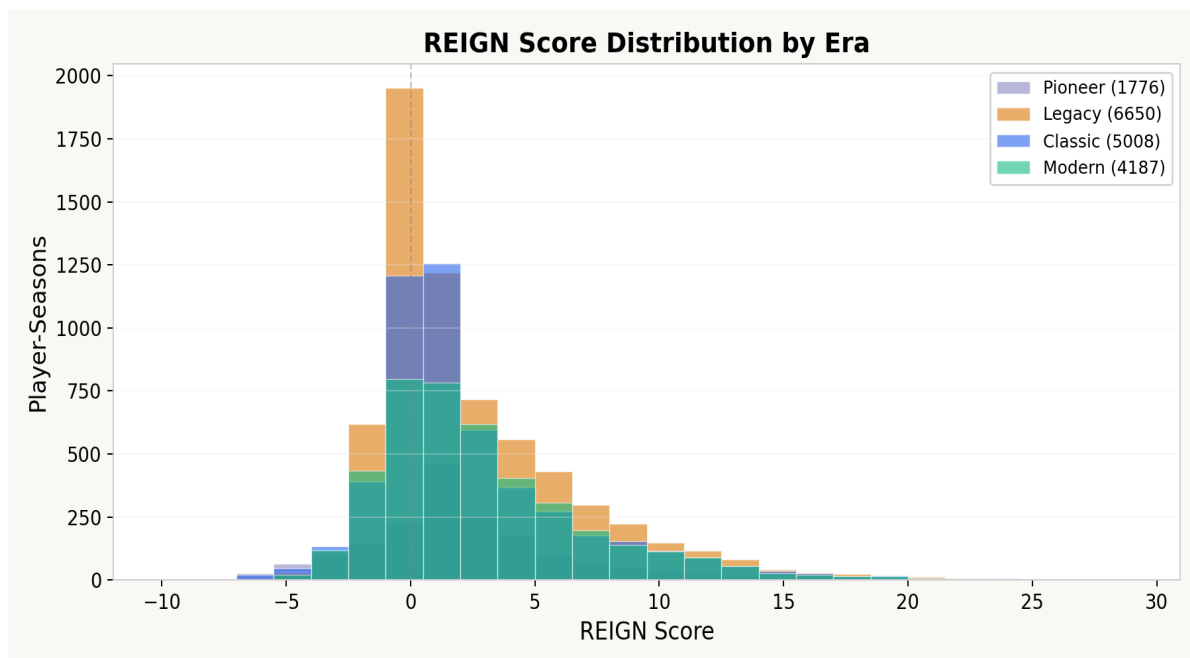


Figure 4: REIGN score distribution by era. All eras show similar spread, confirming cross-era calibration.

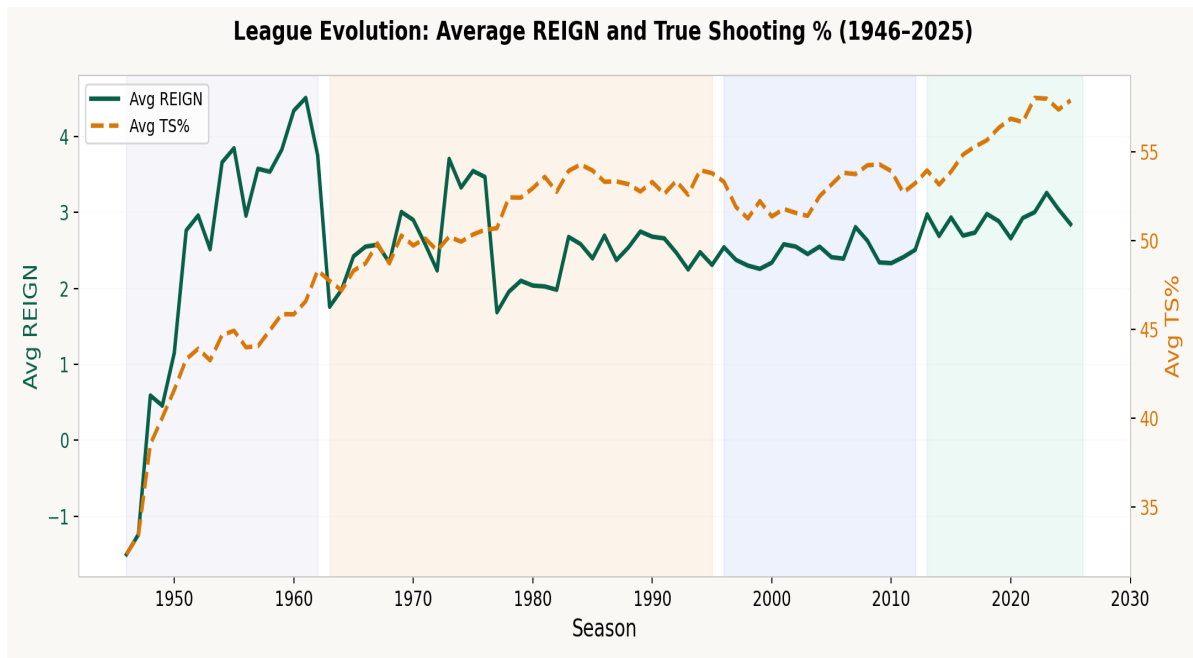


Figure 5: League evolution of average REIGN (green) and True Shooting % (gold) from 1946–2025.

4.4 Offense vs Defense

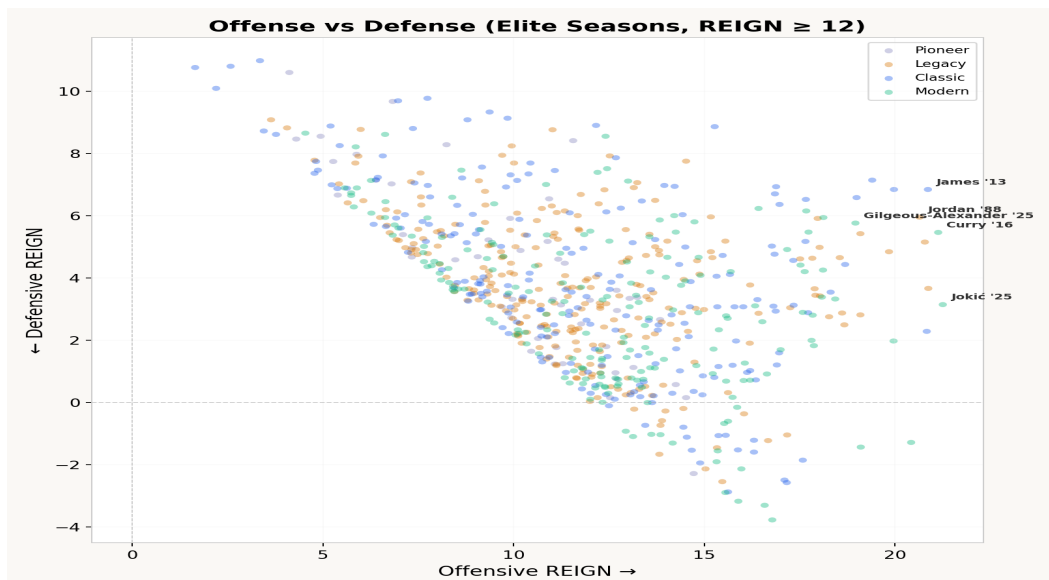


Figure 6: Offensive vs defensive REIGN for elite seasons (≥ 12). Top-right = elite two-way. Top-left = defensive anchor. Bottom-right = pure scorer.

5. Limitations and Future Work

Modern model gap: $R^2=0.790$ vs $0.89-0.92$ for other eras. 2025 season lacks advanced stats.

Blend tradeoff: 60/40 blend caps corrections at ± 4.7 pts. Some residual bias may remain from original model.

Defense: Box score proxies only. Tracking data (contests, deflections) would improve Modern era.

Opponent adjustment: Uses average playoff-team quality, not series-specific matchups.

Future: (1) Series-level opponents; (2) Player tracking for defense; (3) Aging curves; (4) Re-derive when 2025 stats publish.

6. Conclusion

REIGN provides era-specific models with a dampened blend for new data integration, opponent-adjusted playoffs, and full RS/PO decomposition across leaderboards, profiles, comparisons, eras, and visualizations. The interactive platform and formal methodology are publicly available at github.com.

References

- [1] Hollinger, J. (2005). *Pro Basketball Forecast*. Potomac Books.
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Appendix A: Blend Sensitivity

Blend	Max Δ	Mean Δ	>5pt	Selected
70/30	± 3.56	± 0.49	0	
60/40	± 4.74	± 0.66	0	✓
50/50	± 5.93	± 0.82	1	

Table A1: Blend ratio comparison.

Appendix B: Encoding Fix

Basketball Reference serves accented characters as broken UTF-8 byte sequences (e.g., Joki█ → JokiA\x87). 81 international player names required manual mapping. Without this fix, Nikola Joki█'s entire career (3× MVP) had zero advanced statistics, depressing his REIGN by an estimated 3–5 points.